

Note: - for apb address is 32 bit

So for dvsec- [31:19] - 0000

Cisrb – 1000

D2d – 2000

Phy – 3000

for sibeband address is 24 bit

So for dvsec- [23:19] - 0

Cisrb – 1

d2d – 2

phy – 3

UCle DVSEC Register Table

Offset[19:0]	Register Name	Access	Default Value	Register Fields
00h	PCIe Ext Capability Header	RO	32'h0	Next_Capability_Offset[31:20], Revision_ID[19:16], Capability_ID[15:0]
04h	DVSEC Header 1	RO	32'h0	DVSEC_Length[31:16], DVSEC_Vendor_ID[15:0]
08h	{DVSEC Header 2, Capability Descriptor}	RO	32'h0	Reserved1[31:16], DVSEC_ID[15:0]

0Ch	UCle Link Capability	RO	32'h0	Reserved3[31:24], APMW[23:20], Max_Link_Width[19:16], Max_Link_Speed[15:12], Reserved4[11:8], SPMW[7:4], Raw_Format[3:0]
10h	UCle Link Control	RW	32'h0008_0200	Link_Disable[0], Retrain_Link[1], Common_Clock_Config[2], Extended_Sync[3], Equalization_Req[4], Loopback_Enable[5], Reserved1[7:6], Target_Link_Speed[11:8], Enter_Modified_Equalization[12], Auto_Bandwidth_Disable[13], Auto_Bandwidth_Int_En[14], Reserved2[15], Enable_Link[16], Flit_Mode_Enable[17], Training_Init[18], Negotiated_Width_Override[27:19], Reserved3[31:28]
14h	UCle Link Status	RO/RW1C	32'h0	Link_Training[0], DL_Init_Status[1],

				Link_Active[2], Flit_Mode_Active[3], Equalization_In_Progress[4], Negotiated_Link_Speed[7:5], Negotiated_Link_Width[15:8], UCle_State[23:16], Error_Indication[24], Link_Degraded[25], Training_Failure[26], Local_Fault[27], Remote_Fault[28], Reserved1[30:29], Controller_Ready[31]
18h	{Error Notification Control ,Link Event Notification Control}	RW	32'h0	Link Error Notification Interrupt number[31:26], Reserved[25:22], Uncorrectable fatal error detected' UCle Link Error Interrupt enable RP/DSP[21], Uncorrectable fatal error detected' protocol layer based reporting enable[20], Uncorrectable non-fatal error detected' UCle Link Error Interrupt enable[19],

				Uncorrectable non-fatal error detected' protocol layer based reporting enable[18], Correctable error detected' UCle Link Error Interrupt enable[17], Correctable error detected' protocol layer based reporting enable[16] , Link Event Notification Interrupt number[15:11], Reserved1[10:2], 'HW autonomous BW changed' UCle Link Event Interrupt enable[1], 'Link Status changed' UCle Link Event Interrupt enable[0]
1Ch	Register Locator 0 Low	RO	32'h0	Register Block Offset[31:12], Reserved[11:7], Register Block Identifier[6:3], Register BIR[2:0]
20h	Register Locator 0 High	RO	32'h0	Register Block Offset
24h	Register Locator 1 Low	RO	32'h0	Register Block Offset[31:12], Reserved[11:7], Register Block Identifier[6:3], Register BIR[2:0]
28h	Register Locator 1 High	RO	32'h0	Register Block Offset

2Ch	Register Locator 2 Low	RO	32'h0	Register Block Offset[31:12], Reserved[11:7], Register Block Identifier[6:3], Register BIR[2:0]
30h	Register Locator 2 High	RO	32'h0	Register Block Offset
50h	Sideband Mailbox Index (Low)	RW	32'h1e4	Addr[18:0] of Sideband Accesses[31:13], BE[7:0] in sideband [12:5], Opcode[4:0]
54h	Sideband Mailbox Index (High)	RW	32'h0	reserved[31:5], Addr[23:19] of Sideband Accesses [4:0]
58h	Sideband Mailbox Data (Low)	RW	32'h0	For sideband write opcodes, this carries the write data [31:0] to the destination [31:0]
5Ch	Sideband Mailbox Data (High)	RW	32'h0	For sideband write opcodes, this carries the write data [63:32] to the destination.[31:0]
60h	{16'h0, Sideband Mailbox Status, 7'h0, Sideband Mailbox Control}	15-bit is RW 1C RW	{16'h0,8'h0,7'h0,1'b0}	reserved1[31:16], reserved2[15:10], Write/Read status[9:8], reserved3[7:1], Write/Read trigger[0]
64h	Requester ID	RO	32'h0	Reserved[31:24], BDF No for MSIs[23:15]}

6Ch	{16'h0, Associated Port Numbers}	RO	32'h0	{Reserved [31:16], Port Number 2[15:8], Port Number 1[7:0]}
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UCle Switch Register Block (UiSRB) DVSEC Capability

Offset	Register Name	Access	Default Value	Register fields
20'h0	PCI Express Extended Capability Header	RO	{12'h40,4'h1,16'h23}	Next_Capability_Offset[31:20], Revision_ID[19:16], Capability_ID[15:0]
20'h4	Designated Vendor Specific Header 1	RO	{12'h14,4'h0,16'h0}	DVSEC_Length[31:20], revision[19:16], DVSEC_Vendor_ID[15:0]
20'h8	{16'h0, Designated Vendor Specific Header 2}	RO	{16'h0,16'h1}	Reserved[31:16], DVSEC_ID[15:0]
20'hC	CiSRB Base Address (Lower 32 bits)	RO	{20'h0,11'h0,1'b0}	Register Block Offset[31:12] ,Reserved[11:1], Register BIR[0]
20'h10	CiSRB Base Address (Upper 32 bits)	RO	32'h0	Register Block Offset[31:0]

UCle D2D Register Table

Offset	Register Name	Access	Default	Register Fields
00h	UCle Register Block Header Low -0	RO	32'h0	vender id register block [31:16], Vender id [15:0]
04h	UCle Register Block Header Low-1	RO	32'h0	Reserved [31:5] , Vender register block version [3:0]
08h	UCle Register Block Header High-0	RO	32'h0000_2000	Vender block register length [31:0]
0Ch	UCle Register Block Header High-1	RO	32'h0	Reserved [31:0]
10h	Uncorrectable Error Status	RW1CS	32'h0	Reserved[31:6], Invalid Parameter Exchange[5], Sideband Non-Fatal Error Message received[4], Sideband Fatal Error Message received[3], Internal Error[2], Receiver Overflow[1], Adapter Timeout[0]
14h	Uncorrectable Error Mask	RWS	{26'h0, 6'h3f}	Reserved[31:6], Invalid Parameter Exchange Mask[5], Sideband Non-Fatal Error Message received Mask [4],

				Sideband Fatal Error Message received Mask [3], Internal Error Mask [2], Receiver Overflow Mask[1], Adapter Timeout Mask[0]
18h	Uncorrectable Error Severity	RWS	{26'h0, 6'h2f}	Reserved[31:6], Invalid Parameter Exchange Severity [5], Sideband Non-Fatal Error Message received Severity [4], Sideband Fatal Error Message received Severity [3], Internal Error Severity [2], Receiver Overflow Severity [1], Adapter Timeout Severity [0]
1Ch	Correctable Error Status	RW1CS	32'h0	Reserved[31:5], Runtime Link Testing Parity Error[4], Sideband Correctable Error Message received[3], Correctable Internal Error[2], Adapter LSM transition to Retrain[1], CRC Error Detected[0]
20h	Correctable Error Mask	RWS	{27'h0, 5'h1f}	Reserved[31:5], Runtime Link Testing Parity Error Mask[4],

				Sideband Correctable Error Message received Mask [3], Correctable Internal Error Mask [2], Adapter LSM transition to Retrain Mask [1], CRC Error Detected Mask [0]
24h	Header Log 1 Low	ROS	32'h0	Header Log 1[31:0]
28h	Header Log 1 High	ROS	32'h0	Header Log 1[63:32]
2Ch	Header Log 2	ROS	32'h0	Reserved1[31:23], First Fatal Error Indicator [22:18], Flit Format [17:14], Parameter Exchange Successful [13], Reserved [12:11], Adapter LSM id [10], Adapter LSM response type [9:7], Receiver Overflow encoding [6:4], Adapter Timeout encoding [3:0]
30h	Error & Link Testing Control	RW / RW1C	32'h4	Reserved1 [31:18], CRC Injection Busy [17], CRC Injection Count [16:15], CRC Injection Enable [14:13], Reserved 0 [12:10], Parity Feature Nak received [9], Number of 64 Byte Inserts [8:6],

				Runtime Link Testing Rx Enable [5], Runtime Link Testing Tx Enable [4], Remote Register Access Threshold [3:0]
34h	Runtime Link Testing Parity Log 0 Low	RW1C	32'h0	Parity Log for Module 0 [31:0]
38h	Runtime Link Testing Parity Log 0 High	RW1C	32'h0	Parity Log for Module 0 [63:32]
3Ch	Runtime Link Testing Parity Log 1 Low	RW1C	32'h0	Parity Log for Module 1 [31:0]
40h	Runtime Link Testing Parity Log 1 High	RW1C	32'h0	Parity Log for Module 1 [63:32]
44h	Runtime Link Testing Parity Log 2 Low	RW1C	32'h0	Parity Log for Module 2 [31:0]
48h	Runtime Link Testing Parity Log 2 High	RW1C	32'h0	Parity Log for Module 2 [63:32]
4Ch	Runtime Link Testing Parity Log 3 Low	RW1C	32'h0	Parity Log for Module 3 [31:0]
50h	Runtime Link Testing Parity Log 3 High	RW1C	32'h0	Parity Log for Module 3 [63:32]
54h	Advertised Adapter Capability Log Low	RW1C	32'h0	Advertised Adapter Capability [31:0]
58h	Advertised Adapter	RW1C	32'h0	Advertised Adapter Capability [63:32]

	Capability Log High			
5Ch	Finalized Adapter Capability Log Low	RW1C	32'h0	Finalized Adapter Capability [31:0]
60h	Finalized Adapter Capability Log High	RW1C	32'h0	Finalized Adapter Capability [63:32]
64h	Advertised CXL Capability Log Low	RW1C	32'h0	Advertised CXL Capability [31:0]
68h	Advertised CXL Capability Log High	RW1C	32'h0	Advertised CXL Capability [63:32]
6Ch	Finalized CXL Capability Log Low	RW1C	32'h0	Finalized CXL Capability [31:0]
70h	Finalized CXL Capability Log High	RW1C	32'h0	Finalized CXL Capability [63:31]

UCle PHY Register Table

Offset	Register Name	Access	Default	Register Fields
1000h	PHY Capability (global)	RO	32'h0	Reserved2 [31:17], Tightly coupled mode [16], Package type [15], Rx Clock phase support [14:13],

				Rx Clock Mode support [12:11], Reserved1 [10], Supported Tx Vswing encodings [9:5], TX Equalization support[4], Terminated Link[3], Reserved0 [2:0]
1004h	PHY Control (global)	RW	32'h0	Reserved1 [31:9], Force x8 Width Mode in a UCle-S x16 Module [8], Force x32 Width Mode in x64 Module [7], Rx Clock phase support select [6], Rx Clock Mode Select [5], Tx Eq Enable [4], Rx Terminated Control [3], Reserved0 [2:0]
1008h	PHY Status (global)	RO	32'h0	Reserved1 [31:8], Lane Reversal within Module [7], Clock phase Status [6], Clock Mode Status [5], Tx Eq Status [4], Rx Termination Status [3], Reserved0 [2:0]

100Ch	PHY Initialization & Debug	RW	32'h0	Reserved 1 [31:6], Resume Training [5], Reserved 0 [4: 3], Initialization control [2:0]
1010h	Training Setup 1 Reg 0	RW	{5'h0,16'h4,11'h0}	Reserved [31:27], Burst Count [26:11], Training mode [10], Clock Phase control [9:6], Valid Pattern used during training [5: 3], Data pattern used during training [2:0]
1014h	Training Setup 1 Reg 1	RW	{5'h0,16'h4,11'h0}	Reserved [31:27], Burst Count [26:11], Training mode [10], Clock Phase control [9:6], Valid Pattern used during training [5: 3], Data pattern used during training [2:0]
1018h	Training Setup 1 Reg 2	RW	{5'h0,16'h4,11'h0}	Reserved [31:27], Burst Count [26:11], Training mode [10], Clock Phase control [9:6], Valid Pattern used during training [5: 3],

				Data pattern used during training [2:0]
101Ch	Training Setup 1 Reg 3	RW	{5'h0,16'h4,11'h0}	Reserved [31:27], Burst Count [26:11], Training mode [10], Clock Phase control [9:6], Valid Pattern used during training [5: 3], Data pattern used during training [2:0]
1020h	Training Setup 2 Reg0	RW	{16'h4,16'h4}	Iterations [31:16], Idle count [15:0]
1024h	Training Setup 2 Reg1	RW	{16'h4,16'h4}	Iterations [31:16], Idle count [15:0]
1028h	Training Setup 2 Reg2	RW	{16'h4,16'h4}	Iterations [31:16], Idle count [15:0]
102Ch	Training Setup 2 Reg3	RW	{16'h4,16'h4}	Iterations [31:16], Idle count [15:0]
1030h	Training Setup 3 Reg 0 Low	RW	32'h0	Lane mask [31:0]
1034h	Training Setup 3 Reg 0 High	RW	32'h0	Lane mask [63:32]
1038h	Training Setup 3 Reg 1 Low	RW	32'h0	Lane mask [31:0]
103Ch	Training Setup 3 Reg 1 High	RW	32'h0	Lane mask [63:32]
1040h	Training Setup 3 Reg 2 Low	RW	32'h0	Lane mask [31:0]

1044h	Training Setup 3 Reg 2 High	RW	32'h0	Lane mask [63:32]
1048h	Training Setup 3 Reg 3 Low	RW	32'h0	Lane mask [31:0]
104Ch	Training Setup 3 Reg 3 High	RW	32'h0	Lane mask [63:32]
1050h	Training Setup 4 Reg 0	RW	32'h0	Max error Threshold in aggregate comparison [31:16], Max error Threshold in per Lane comparison [15:4], Repair Lane mask [3:0]
1054h	Training Setup 4 Reg 1	RW	32'h0	Max error Threshold in aggregate comparison [31:16], Max error Threshold in per Lane comparison [15:4], Repair Lane mask [3:0]
1058h	Training Setup 4 Reg 2	RW	32'h0	Max error Threshold in aggregate comparison [31:16], Max error Threshold in per Lane comparison [15:4], Repair Lane mask [3:0]
105Ch	Training Setup 4 Reg 3	RW	32'h0	Max error Threshold in aggregate comparison [31:16], Max error Threshold in per Lane comparison [15:4], Repair Lane mask [3:0]
1060h	Current Lane Map Module 0 Low	RW	32'h0	Current Rx Lane map (CLM) for Module-0 [31:0]

1064h	Current Lane Map Module 0 High	RW	32'h0	Current Rx Lane map (CLM) for Module-0 [63: 32]
1068h	Current Lane Map Module 1 Low	RW	32'h0	Current Rx Lane map (CLM) for Module-1 [31:0]
106Ch	Current Lane Map Module 1 High	RW	32'h0	Current Rx Lane map (CLM) for Module-1 [63: 32]
1070h	Current Lane Map Module 2 Low	RW	32'h0	Current Rx Lane map (CLM) for Module-2 [31:0]
1074h	Current Lane Map Module 2 High	RW	32'h0	Current Rx Lane map (CLM) for Module-2 [63: 32]
1078h	Current Lane Map Module 3 Low	RW	32'h0	Current Rx Lane map (CLM) for Module-3 [31:0]
107Ch	Current Lane Map Module 3 High	RW	32'h0	Current Rx Lane map (CLM) for Module-3 [63: 32]
1080h	Error Log 0 Reg 0	ROS	32'h0	State (N-2) [31:24], State (N-1) [23:16], Reserved [15:10], Width Degrade [9], Lane Reversal [8], State N [7:0]
1084h	Error Log 0 Reg 1	ROS	32'h0	State (N-2) [31:24], State (N-1) [23:16], Reserved [15:10], Width Degrade [9], Lane Reversal [8],

				State N [7:0]
1088h	Error Log 0 Reg 2	ROS	32'h0	State (N-2) [31:24], State (N-1) [23:16], Reserved [15:10], Width Degrade [9], Lane Reversal [8], State N [7:0]
108Ch	Error Log 0 Reg 3	ROS	32'h0	State (N-2) [31:24], State (N-1) [23:16], Reserved [15:10], Width Degrade [9], Lane Reversal [8], State N [7:0]
1090h	Error Log 1 Reg 0	RW1CS	32'h0	Reserved [31:12], Internal Error [11], Remote LinkError received [10], Sideband Timeout Occurred [9], State Timeout Occurred [8], State (N-3) [7:0]
1094h	Error Log 1 Reg 1	RW1CS	32'h0	Reserved [31:12], Internal Error [11], Remote LinkError received [10],

				Sideband Timeout Occurred [9], State Timeout Occurred [8], State (N-3) [7:0]
1098h	Error Log 1 Reg 2	RW1CS	32'h0	Reserved [31:12], Internal Error [11], Remote LinkError received [10], Sideband Timeout Occurred [9], State Timeout Occurred [8], State (N-3) [7:0]
109Ch	Error Log 1 Reg 3	RW1CS	32'h0	Reserved [31:12], Internal Error [11], Remote LinkError received [10], Sideband Timeout Occurred [9], State Timeout Occurred [8], State (N-3) [7:0]
1100h	Runtime Link Test Control Low	RW	32'h0	Module 3 Lane repair id [31:29], Module 2 Lane repair id [28:22], Module 1 Lane repair id [21:15], Module 0 Lane repair id [14:8], Inject Stuck-at fault [7],

				Start [6], Apply Module 3 Lane Repair [5], Apply Module 2 Lane Repair [4], Apply Module 1 Lane Repair [3], Apply Module 0 Lane Repair [2], Reserved [1:0]
1104h	Runtime Link Test Control High	RW	32'h0	Reserved [31:4], Module 3 Lane repair id 35:32 [3:0]
1108h	Runtime Link Test Status	RO	32'h0	Reserved [31:1], Busy [0]
110Ch	Mainband Data Repair Reg0 Low	RO	32'hFFFF_FFFF	Repair Address for TRD_P[3] [31:24], Repair Address for TRD_P[2] [23:16], Repair Address for TRD_P[1] [15:8], Repair Address for TRD_P[0] [7:0]
1110h	Mainband Data Repair Reg0 High	RO	32'hFFFF_FFFF	Repair Address for RRD_P[3] [31:24], Repair Address for RRD_P[2] [23:16], Repair Address for RRD_P[1] [15:8],

				Repair Address for RRD_P[0] [7:0]
1114h	Mainband Data Repair Reg1 Low	RO	32'hFFFF_FFFF	Repair Address for TRD_P[3] [31:24],Repair Address for TRD_P[2] [23:16], Repair Address for TRD_P[1] [15:8], Repair Address for TRD_P[0] [7:0]
1118h	Mainband Data Repair Reg1 High	RO	32'hFFFF_FFFF	Repair Address for RRD_P[3] [31:24], Repair Address for RRD_P[2] [23:16], Repair Address for RRD_P[1] [15:8], Repair Address for RRD_P[0] [7:0]
111Ch	Mainband Data Repair Reg 2 Low	RO	32'hFFFF_FFFF	Repair Address for TRD_P[3] [31:24],Repair Address for TRD_P[2] [23:16], Repair Address for TRD_P[1] [15:8], Repair Address for TRD_P[0] [7:0]
1120h	Mainband Data Repair Reg2 High	RO	32'hFFFF_FFFF	Repair Address for RRD_P[3] [31:24], Repair Address for RRD_P[2] [23:16], Repair Address for RRD_P[1] [15:8],

				Repair Address for RRD_P[0] [7:0]
1124h	Mainband Data Repair Reg3 Low	RO	32'hFFFF_FFFF	Repair Address for TRD_P[3] [31:24],Repair Address for TRD_P[2] [23:16], Repair Address for TRD_P[1] [15:8], Repair Address for TRD_P[0] [7:0]
1128h	Mainband Data Repair Reg3 High	RO	32'hFFFF_FFFF	Repair Address for RRD_P[3] [31:24], Repair Address for RRD_P[2] [23:16], Repair Address for RRD_P[1] [15:8], Repair Address for RRD_P[0] [7:0]
1134h	Clock, Track, Valid and Sideband Repair Reg0	RO	32'h0000_0fff	Reserved [31:24], Repair Address for Sideband Receiver [23:20], Repair Address for Sideband Transmitter [19:16], Reserved [15:12], Repair Address for RRDVLD_P [11:10], Repair Address for TRDVLD_P [9:8], Repair Address for RRDCK_P [7:4],

				Repair Address for TRDCK_P [3:0]
1138h	Clock, Track, Valid and Sideband Repair Reg1	RO	32'h0000_0fff	Reserved [31:24], Repair Address for Sideband Receiver [23:20], Repair Address for Sideband Transmitter [19:16], Reserved [15:12], Repair Address for RRDVLD_P [11:10], Repair Address for TRDVLD_P [9:8], Repair Address for RRDCK_P [7:4], Repair Address for TRDCK_P [3:0]
113Ch	Clock, Track, Valid and Sideband Repair Reg2	RO	32'h0000_0fff	Reserved [31:24], Repair Address for Sideband Receiver [23:20], Repair Address for Sideband Transmitter [19:16], Reserved [15:12], Repair Address for RRDVLD_P [11:10], Repair Address for TRDVLD_P [9:8], Repair Address for RRDCK_P [7:4],

				Repair Address for TRDCK_P [3:0]
1140h	Clock, Track, Valid and Sideband Repair Reg3	RO	32'h0000_0fff	Reserved [31:24], Repair Address for Sideband Receiver [23:20], Repair Address for Sideband Transmitter [19:16], Reserved [15:12], Repair Address for RRDVLD_P [11:10], Repair Address for TRDVLD_P [9:8], Repair Address for RRDCK_P [7:4], Repair Address for TRDCK_P [3:0]